Lab 9: Inheritance Genetics — Genetics Problem Solving

BIOL-1

Name: ______________________________ **Date:** ______________________________

Overview

Gregor Mendel discovered that traits are inherited in predictable patterns. By crossing pea plants and counting offspring, he uncovered the rules of genetics — rules that apply to all living things, including humans. In this lab, you will use **Punnett squares** to predict the offspring of genetic crosses, explore **blood types**, analyze **sex-linked traits**, and interpret a **pedigree**.

Learning Objectives

By the end of this lab, you will be able to:

1. Use Punnett squares to predict genotypic and phenotypic ratios.
2. Solve problems involving incomplete dominance and codominance.
3. Determine blood type inheritance using the ABO allele system.
4. Predict outcomes of X-linked trait crosses.
5. Analyze a pedigree to determine the pattern of inheritance.

Materials

- This worksheet
- Scratch paper for additional Punnett squares

Part 1: Monohybrid Crosses

In pea plants, Tall (T) is dominant over Short (t). A heterozygous plant (Tt) looks tall but carries the short allele.

1. Cross a heterozygous tall plant (Tt) with a short plant (tt). Fill in the Punnett square:

	t	t
T		
t		

- Genotypic ratio:
- Phenotypic ratio:
- What percentage of offspring will be short?

2. Cross two heterozygous tall plants (Tt × Tt). Fill in the Punnett square:

	T	t
T		
t		

- Genotypic ratio:
- Phenotypic ratio:
- What is the chance of a homozygous recessive (tt) offspring?

Part 2: Incomplete Dominance

In incomplete dominance, the heterozygote is a BLEND. Neither allele is fully dominant.

Example: In snapdragons, Red (RR) × White (R′R′) = Pink (RR′)

3. Cross two pink snapdragons (RR′ × RR′). Fill in the Punnett square:

	R	R′
R		
R′		

- Phenotypic ratio: ***Red :*** Pink : ____ White

4. How is incomplete dominance different from complete dominance?

Part 3: ABO Blood Types

The ABO blood system demonstrates both codominance (I^A and I^B are codominant) and multiple alleles (three alleles: I^A, I^B, i). This system determines which antigens are on your red blood cells and is essential for safe blood transfusions.

5. Fill in the genotypes for each blood type:

Blood Type	Possible Genotype(s)	Antigens on Red Blood Cells
Type A		A antigen
Type B		B antigen
Type AB		Both A and B
Type O		Neither

6. A mother is Type A (I^Ai) and a father is Type B (I^Bi). Fill in the Punnett square:

	I^B	i
I^A		
i		

7. What blood types are possible in their children?

8. Can they have a child with Type O blood? Explain using the Punnett square.

Part 4: Sex-Linked Traits

Color blindness is caused by a recessive allele on the X chromosome. Because males have only one X (XY), a single recessive allele is enough to cause the condition. Females (XX) need two copies.

Normal vision = X^B. Color blind = X^b.

9. A carrier mother (X^B X^b) and a normal-vision father (X^B Y) have children. Fill in the Punnett square:

	X^B	Y
X^B		
X^b		

10. What fraction of their sons will be colorblind?

11. What fraction of their daughters will be carriers?

12. Explain in your own words why X-linked recessive traits are more common in males.

Part 5: Pedigree Analysis

A pedigree is a family tree showing who in a family is affected by a genetic trait. Circles = females, Squares = males, Filled shapes = affected individuals.

Pedigree: Two unaffected parents (Generation I) have four children (Generation II): an affected son, an unaffected son, an unaffected carrier daughter, and an unaffected daughter. The carrier daughter later marries an unaffected man and has an affected son (Generation III).

13. Is this trait dominant or recessive? How do you know?

14. Is this trait autosomal or X-linked? What evidence supports your answer?

15. What are the genotypes of the Generation I parents?

Part 6: Polygenic Traits

Not all traits follow simple Mendelian patterns. Many human traits — like height, skin color, and eye color — are controlled by MANY genes working together.

16. What is a polygenic trait? Name two examples.

17. Mendel's pea plants were either tall OR short (two categories). Human height shows a continuous range from very short to very tall. Why is human height distributed as a bell curve instead of two distinct categories?

18. A student says: "Genes determine everything about you — environment doesn't matter." Do you agree or disagree? Give a specific example to support your answer.